Assignment_6 矩阵、贪心

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1. 题目

`M21554`：排队做实验

greedy, http://cs101.openjudge.cn/pctbook/M21554/

```
n = int(input())
students = list(map(int, input().split()))
initial = dict(enumerate(students, 1))
final = dict(sorted(initial.items(), key=lambda x: x[1]))

total_waiting_time = 0
k = 1
for i, student in final.items():
    total_waiting_time += student * (n - k)
    k += 1
print(*final.keys())
average_waiting_time = total_waiting_time / n
print(f"{average_waiting_time:.2f}")
```

or

```
n = int(input())
ini_students = list(map(int, input().split()))
students = []
for i in range(n):
    students.append((ini_students[i], i))
students.sort()

total_waiting_time = 0
order = []
for i in range(n):
    order.append(str(students[i][1] + 1))
    total_waiting_time += students[i][0] * (n - i - 1)

print(' '.join(order))
print(f"{total_waiting_time / n:.2f}")
```

代码运行截图 （至少包含有"Accepted"）



#53509831提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
n = int(input())
students = list(map(int, input().split()))
initial = dict(enumerate(students, 1))
final = dict(sorted(initial.items(), key=lambda x: x[1]))

total_waiting_time = 0
k = 1
for i, student in final.items():
    total_waiting_time += student * (n - k)
    k += 1
print(*final.keys())
average_waiting_time = total_waiting_time / n
print(f"{average_waiting_time:.2f}")
```

基本信息

#: 53509831
题目: M21554
提交人: 26n2600011560
内存: 3648kB
时间: 29ms
语言: Python3
提交时间: 2026-10-17 16:43:36

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English 帮助 关于

M18211：军备竞赛

greedy, two pointers, http://cs101.openjudge.cn/pctbook/M18211

```
p = int(input())
weapons = list(map(int, input().split()))
weapons.sort()

l = 0
r = len(weapons) - 1
cur = 0
ans = 0
while l <= r:
    if p - weapons[l] < 0:
        if cur >= 1:
            p += weapons[r]
            r -= 1
            cur -= 1
        else:
            break
    else:
        p -= weapons[l]
        l += 1
        cur += 1
        ans = cur

print(ans)
```

代码运行截图 （至少包含有"Accepted"）



CS101 / 计算思维算法实践

题目 排名 状态 提问

#53454019提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
p = int(input())
weapons = list(map(int, input().split()))
weapons.sort()

l = 0
r = len(weapons) - 1
cur = 0
ans = 0
while l <= r:
    if p - weapons[l] < 0:
        if cur >= 1:
            p += weapons[r]
            r -= 1
            cur -= 1
        else:
            break
    else:
        p -= weapons[l]
        l += 1
        cur += 1
        ans = cur

print(ans)
```

基本信息

#: 53454019
题目: M18211
提交人: 26n2600011560
内存: 3636kB
时间: 23ms
语言: Python3
提交时间: 2026-10-13 16:38:42

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English 帮助 关于

E23555：节省存储的矩阵乘法

implementation, matrices, http://cs101.openjudge.cn/pctbook/E23555

```
n, m1, m2 = map(int, input().split())
A = [[0] * n for _ in range(n)]
B = [[0] * n for _ in range(n)]
for _ in range(m1):
    r, c, num = map(int, input().split())
    A[r][c] = num
for _ in range(m2):
    r, c, num = map(int, input().split())
    B[r][c] = num

C = [[0] * n for _ in range(n)]
for i in range(n):
    for j in range(n):
        for k in range(n):
            C[i][j] += A[i][k] * B[k][j]

for r in range(n):
    for c in range(n):
```

```
            if C[r][c] != 0:
                print(r, c, C[r][c])
```

or

```
n, m1, m2 = map(int, input().split())
A = [[0] * n for _ in range(n)]
B = [[0] * n for _ in range(n)]
for _ in range(m1):
    r, c, num = map(int, input().split())
    A[r][c] = num
for _ in range(m2):
    r, c, num = map(int, input().split())
    B[r][c] = num

C = [[0] * n for _ in range(n)]
for i in range(n):
    A_i = A[i]
    C_i = C[i]
    # 提到内层循环外，减少索引
    for j in range(n):
        if A_i[j] != 0:  # 跳过0，常数优化
            B_j, v = B[j], A_i[j]
            for t in range(n):
                C_i[t] += v * B_j[t]

for r in range(n):
    for c in range(n):
        if C[r][c] != 0:
            print(r, c, C[r][c])
```

代码运行截图 （至少包含有"Accepted"）



CS101 / 计算思维算法实践

题目 排名 状态 提问

#53455303提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
n, m1, m2 = map(int, input().split())
A = [[0] * n for _ in range(n)]
B = [[0] * n for _ in range(n)]
for _ in range(m1):
    r, c, num = map(int, input().split())
    A[r][c] = num
for _ in range(m2):
    r, c, num = map(int, input().split())
    B[r][c] = num

C = [[0] * n for _ in range(n)]
for i in range(n):
    for j in range(n):
        for k in range(n):
            C[i][j] += A[i][k] * B[k][j]

for r in range(n):
    for c in range(n):
        if C[r][c] != 0:
            print(r, c, C[r][c])
```

基本信息

#: 53455303
题目: E23555
提交人: 26n2600011560
内存: 3776kB
时间: 34ms
语言: Python3
提交时间: 2026-10-13 17:49:53

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English 帮助 关于

M12558：岛屿周长

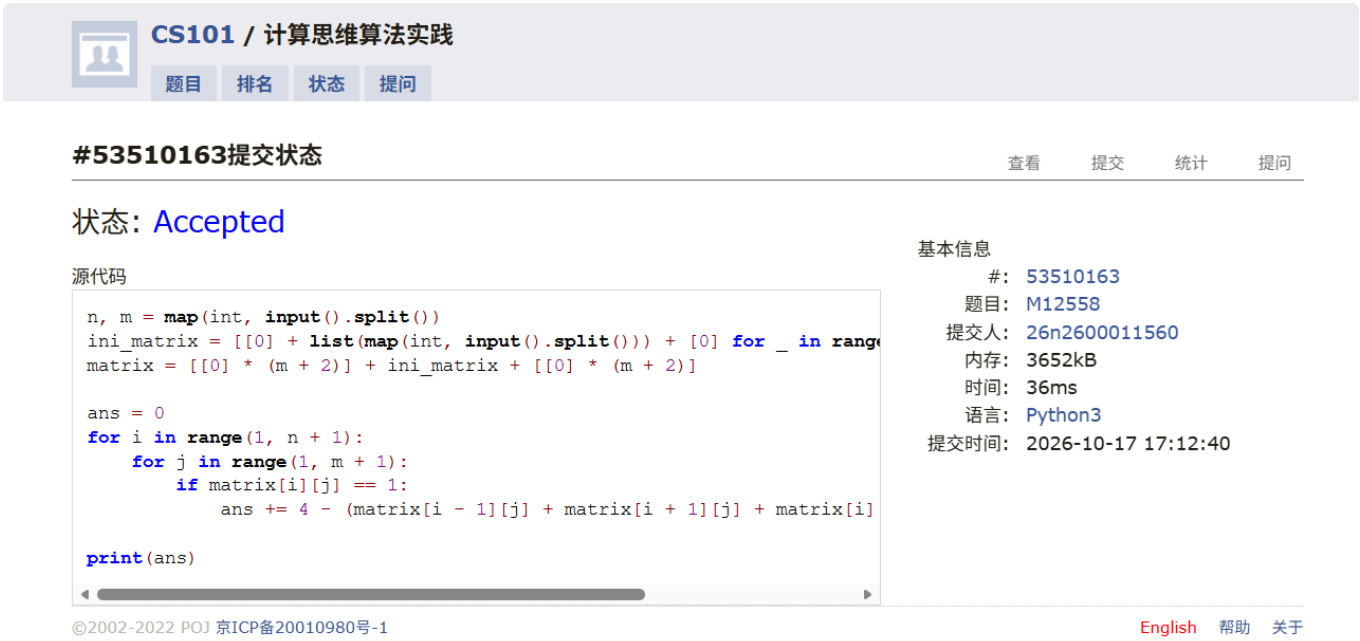
matrices, http://cs101.openjudge.cn/pctbook/M12558

```
n, m = map(int, input().split())
ini_matrix = [[0] + list(map(int, input().split())) + [0] for _ in range(n)]
matrix = [[0] * (m + 2)] + ini_matrix + [[0] * (m + 2)]

ans = 0
for i in range(1, n + 1):
    for j in range(1, m + 1):
        if matrix[i][j] == 1:
            ans += 4 - (matrix[i - 1][j] + matrix[i + 1][j] + matrix[i][j - 1]
+ matrix[i][j + 1])

print(ans)
```

代码运行截图 （至少包含有"Accepted"）



M01328 : Radar Installation

greedy, http://cs101.openjudge.cn/practice/01328/

（这个greedy办法不完全是自己想出来的）

- 若 $\exists 1 \leq i \geq n, \text{s.t.} y_i > d$，显然无解；
- 否则我们考虑其中某个坐标为 (x_i, y_i) 的岛屿，令 $r_i = \sqrt{d^2 - y_i^2}$，此岛屿允许的雷达位置便为 $[x_i - r_i, x_i + r_i]$，接下来的问题就通通都在于看这些区间怎么样相互重叠.....

```
import sys
import math


def main():
    data = sys.stdin.read().split()
    if not data:
        return

    idx = 0
    case_num = 0
    while idx < len(data):
        n = int(data[idx])
        d = int(data[idx + 1])
        idx += 2
        case_num += 1

        if n == 0 and d == 0:
```

```
            break

        islands = []
        max_y = 0
        for _ in range(n):
            x = int(data[idx])
            y = int(data[idx + 1])
            islands.append((x, y))
            if y > max_y:
                max_y = y
            idx += 2

        if max_y > d:
            print(f"Case {case_num}: -1")
            continue

        ranges = []
        for x, y in islands:
            offset = math.sqrt(d * d - y * y)
            ranges.append((x - offset, x + offset))

        # 通过.sort()使临近区间在一起，方便了后续邻两项间的判断
        ranges.sort(key=lambda item: item[1])

        ans = 1
        last_radar_pos = ranges[0][1]

        for i in range(1, n):
            l, r = ranges[i]
            if l > last_radar_pos:
                ans += 1
                last_radar_pos = r
        print(f"Case {case_num}: {ans}")


if __name__ == "__main__":
    main()
```

or

```
import sys
import math


def main():
```

```
    data = sys.stdin.read().split()
    if not data:
        return

    idx = 0
    case_num = 0
    while idx < len(data):
        n = int(data[idx])
        d = int(data[idx + 1])
        idx += 2
        case_num += 1

        if n == 0 and d == 0:
            break

        islands = []
        max_y = 0
        for _ in range(n):
            x = int(data[idx])
            y = int(data[idx + 1])
            islands.append((x, y))
            if y > max_y:
                max_y = y
            idx += 2

        if max_y > d:
            print(f"Case {case_num}: -1")
            continue

        ranges = []
        for x, y in islands:
            offset = math.sqrt(d * d - y * y)
            ranges.append((x - offset, x + offset))

        # 通过.sort()使临近区间在一起，方便了后续邻两项间的判断
        ranges.sort(key=lambda item: item[0], reverse=True)

        ans = 1
        last_radar_pos = ranges[0][0]

        for i in range(1, n):
            l, r = ranges[i]
            if r < last_radar_pos:
                ans += 1
                last_radar_pos = l
        print(f"Case {case_num}: {ans}")
```

```
if __name__ == "__main__":
    main()
```

代码运行截图 （至少包含有"Accepted"）

#53512172提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
import sys
import math


def main():
    data = sys.stdin.read().split()
    if not data:
        return

    idx = 0
    case_num = 0
    while idx < len(data):
        n = int(data[idx])
        d = int(data[idx + 1])
        idx += 2
        case_num += 1

        if n == 0 and d == 0:
            break

        islands = []
        max_y = 0
        for _ in range(n):
            x = int(data[idx])
            y = int(data[idx + 1])
            islands.append((x, y))
            if y > max_y:
                max_y = y
            idx += 2

        if max_y > d:
            print(f"Case {case_num}: -1")
```

基本信息

#: 53512172
题目: 01328
提交人: 26n2600011560
内存: 4964kB
时间: 35ms
语言: Python3
提交时间: 2026-10-17 20:39:17

545C : Woodcutters

dp, greedy, 1500, https://codeforces.com/problemset/problem/545/C

```
n = int(input())
trees = [tuple(map(int, input().split())) for _ in range(n)]

if n <= 2:
    print(n)
else:
    ans = 2
    last_r = trees[0][0]

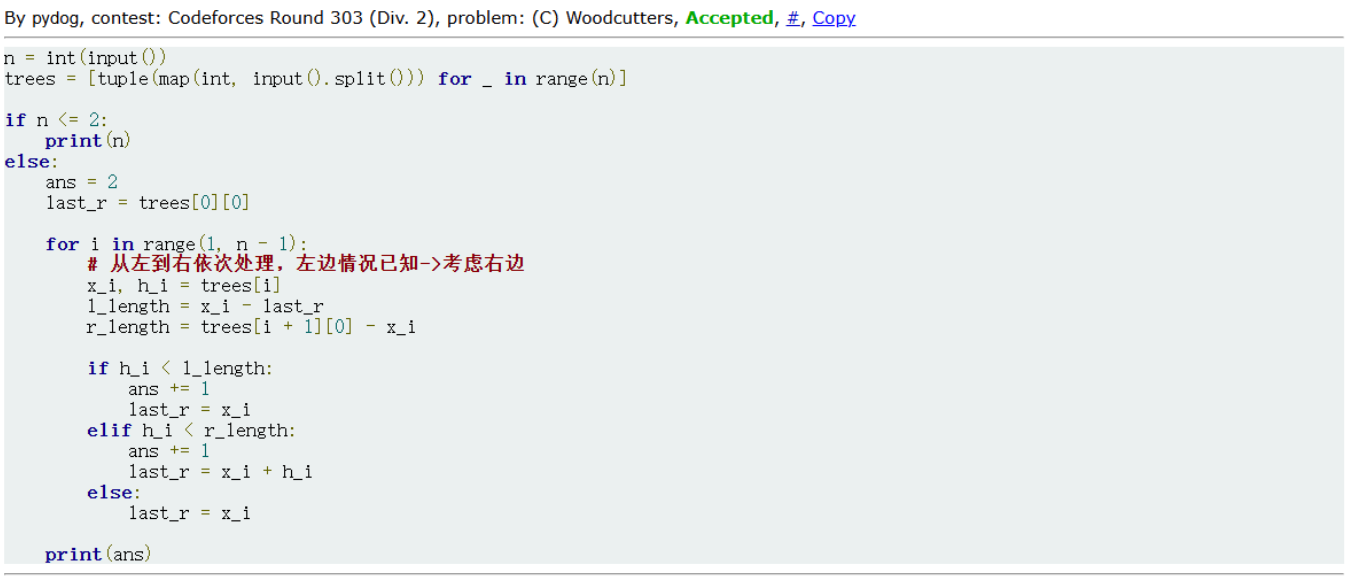
    for i in range(1, n - 1):
```

```
        # 从左到右依次处理，左边情况已知->考虑右边
        x_i, h_i = trees[i]
        l_length = x_i - last_r
        r_length = trees[i + 1][0] - x_i

        if h_i < l_length:
            ans += 1
            last_r = x_i
        elif h_i < r_length:
            ans += 1
            last_r = x_i + h_i
        else:
            last_r = x_i

    print(ans)
```

代码运行截图 （至少包含有"Accepted"）



By pydog, contest: Codeforces Round 303 (Div. 2), problem: (C) Woodcutters, **Accepted**, #, Copy

```
n = int(input())
trees = [tuple(map(int, input().split())) for _ in range(n)]

if n <= 2:
    print(n)
else:
    ans = 2
    last_r = trees[0][0]

    for i in range(1, n - 1):
        # 从左到右依次处理，左边情况已知->考虑右边
        x_i, h_i = trees[i]
        l_length = x_i - last_r
        r_length = trees[i + 1][0] - x_i

        if h_i < l_length:
            ans += 1
            last_r = x_i
        elif h_i < r_length:
            ans += 1
            last_r = x_i + h_i
        else:
            last_r = x_i

    print(ans)
```

2. 学习总结和收获

这次作业很有趣嗷...

- 喜：除了T5都独立AC了（第2题尤其值得高兴）
- 悲：T5真没独立想清楚；T3的优化办法绕了好长时间

P.S. 最近忙别的都没做上每日一题，忏悔！